Shifted Exponential Distribution

1 Definition

A continuous RV X is said to follow a shifted exponential distribution if it's PDF is given by,

$$f(x) = \begin{cases} \frac{1}{\sigma} e^{-\frac{x-\mu}{\sigma}} & , x \geqslant \mu \\ 0 & , \text{ otherwise} \end{cases}$$

2 Moments

$$\begin{aligned} E(X-\mu)^r &= \int_{-\infty}^{\infty} (x-\mu)^r f_X(x) dx \quad \frac{x-\mu}{\sigma} = z, dx = \sigma dz \\ &= \int_{\mu}^{\infty} (x-\mu)^n \frac{1}{\sigma} e^{-\left(\frac{x-\mu}{\sigma}\right)} dx \\ &= \frac{1}{\sigma} \int_0^{\infty} (\sigma z)^r \frac{1}{\sigma} e^{-z} \sigma dz \\ &= \sigma^r \int_0^{\infty} z^{r+1-1} e^{-z} dz \\ &= \sigma^r \Gamma r + 1 \end{aligned}$$

$\Rightarrow E(X-\mu) = \sigma \Rightarrow E(X) = \mu + \sigma$

$$\begin{aligned} & E(X-\mu)^2 = \sigma^2 \\ \Rightarrow & E\left(X^2\right) + \mu^2 - 2\mu E(X) = 2\sigma^2 \\ & E\left(X^2\right) + \mu^2 - 2\mu(\mu+\sigma) = 2\sigma^2 \\ & E\left(X^2\right) + \mu^2 - 2\mu^2 - 2\mu\sigma = 2\sigma^2 \\ & E\left(X^2\right) = 2\sigma^2 + 2\mu\sigma + \mu^2 \end{aligned}$$

$$\begin{aligned} V(X) &= 2\sigma^2 + 2\mu\sigma + \mu^2 - (\mu+\sigma)^2 \\ &= 2\sigma^2 + 2\mu\sigma + \mu^2 - \mu^2 - \sigma^2 - 2\mu\sigma \\ &= \sigma^2 \end{aligned}$$

3 CDF

$$\begin{aligned} F_X(x) &= \int_{-\infty}^{x} f(t) dt \\ &= \begin{cases} 0 & x \leq \mu \\ 1 - e^{-\left(\frac{x-\mu}{\sigma}\right)} & x > \mu \end{cases} \end{aligned}$$

4 Quintiles

$$\begin{aligned} p &= f(\xi_p) \\ \Rightarrow 1 - e^{-\left(\frac{\xi_p - \mu}{\sigma}\right)} &= p \Rightarrow \xi_p = \mu - \sigma \ln(1-p) \\ \xi_{1/2} &= \mu + \sigma \ln 2 \end{aligned}$$

5 MD about Median

$$\begin{aligned} E|X - \mu - \sigma \ln 2| &= \int_{\mu}^{\infty} |x - \mu - \sigma \ln 2| \frac{1}{\sigma} e^{-\frac{x-\mu}{\sigma}} dx \\ &= \int_{0}^{\infty} |y - \sigma \ln 2| \frac{1}{\sigma} e^{-y/\sigma} dy \\ &= \sigma \ln 2. \end{aligned}$$

So, $X \sim$ shifted $\text{Exp}(\mu, \sigma) \Rightarrow Y = X - \mu \sim \text{Exp}(\sigma)$

Remark: The Lack of Memory property is not satisfied by shifted Exponential Distribution.

$$\begin{aligned} P[X > a + b \mid X > a] &= \frac{1 - F_X(a+b)}{1 - F_X(a)} \\ &= \frac{e^{-\left(\frac{a+b-\mu}{\sigma}\right)}}{e^{-\left(\frac{a-\mu}{\sigma}\right)}} = e^{-b/\sigma} + e^{-\left(\frac{b-\mu}{\sigma}\right)} \\ &\neq P[X > b] \end{aligned}$$

1. If $X_1, X_2, \ldots X_n \overset{\text{iid}}{\sim}$ shifted $\text{Exp}(\mu, \sigma)$. Then, find the distribution of $\min(X_1, X_2 \cdots X_n) = Y$

$$\begin{aligned} F_Y(y) &= p[Y \leq y] \\ &= p[\min(X_1, X_2, \ldots X_n) \leqslant y] \\ &= 1 - p[\min(X_1, X_2, \ldots X_n) > y] \\ &= 1 - p[X_1 > y_1, X_2 > y_1 \ldots X_n > y] \\ &= 1 - [p(X_1 > y)]^n \\ &= 1 - [1 - p(X_1 \leqslant y)]^n \\ &= 1 - \left[1 - 1 + e^{-\left(\frac{y-\mu}{\sigma}\right)}\right]^n = 1 - e^{-\left(\frac{y-\mu}{\sigma}\right)^n} \end{aligned}$$

2. $X_1, X_2, \ldots X_n \overset{iid}{\sim} \text{Exp}(\theta)$.

$$\begin{aligned} Y &= \min(X_1, X_2, \ldots X_n) \\ F_Y(y) = P[Y \leqslant y] &= P[\min(X_1, X_2, \ldots X_n) \leqslant y] \\ &= 1 - P[\min(X_1, X_2, \ldots, X_n) > y] \\ &= 1 - P[X_1 > y_1, X_2 > y, \ldots, X_n > y] = \quad 1 - [P(X_1 > y)]^n \\ &= 1 - [1 - P(X_1 \leqslant y)]^n \\ &= 1 - \left[1 - 1 + e^{-y/\theta}\right]^n = 1 - e^{-\frac{ny}{\theta}} \end{aligned}$$